TECHNIJIAN
technology as a solution

My AI — Manufacturing Edition

PRECISION MANUFACTURING AI

Aerospace • Defense • ITAR • AS9100 • ISO 9001

25+ Years Exp | **ITAR** Aware | **~70%** Faster Quotes | **100%** Hands-On

AI purpose-built for precision manufacturing — intelligent estimating, smart fabrication, quality AI, and compliance documentation.

MANUFACTURING AI SERVICES

Estimating & Quoting Intelligence

- ✓ Drawing intake & GD&T feature extraction
- ✓ Automated BOM generation & material calcs
- ✓ Historical quote benchmarking & win-rate analysis
- ✓ Complexity scoring & margin modeling
- ✓ Quote turnaround: days → hours

Smart Fabrication & Production

- ✓ Production scheduling optimization
- ✓ Machine utilization & OEE intelligence
- ✓ Work-in-progress tracking & delay prediction
- ✓ Setup time reduction via AI sequencing
- ✓ Shop-floor data harmonization

Engineering & Quality AI

- ✓ Drawing/spec review with AI-assisted markup
- ✓ Quality escape pattern detection & root cause
- ✓ NCR summarization & trend analysis
- ✓ Supplier quality scoring & risk ranking
- ✓ Engineering knowledge preservation (retirees)

Compliance & Documentation

- ✓ AS9100, ISO 9001, ITAR-compliant architectures
- ✓ PPAP, FAI, CoC automated package generation
- ✓ Air-gapped & US-only deployment options
- ✓ Controlled document retrieval with RBAC
- ✓ CUI / NIST SP 800-171 alignment

HOW WE ENGAGE

1. **Discover** — Map workflows, pain points & AI opportunities
2. **Design** — Architect solutions for goals & compliance
3. **Implement** — Build, test & deploy with hands-on engineering
4. **Operate** — Operationalize with training, governance & support

MANUFACTURING AI USE CASES

Use Case	Focus Area	Impact
Estimating Intelligence	Drawing analysis, BOM generation, ITAR/AS9100 docs	Days→Hrs
Intelligent Quoting	Spec review, complexity scoring, production scheduling	Multi-process
Drawing & Spec AI	Extract GD&T, generate BOMs, calculate material	Auto-extract
Compliance Doc AI	PPAP / FAI / CoC auto-generation, audit-ready	50%+ less effort
Knowledge Preservation	Capture senior estimator/engineer expertise in AI	Tribal→RAG

WHY TECHNIJIAN

- **Manufacturing domain depth:** Drawings, GD&T, BOMs, PPAP, FAI — not generic enterprise AI
- **ITAR & AS9100 aware:** Air-gapped and US-only architectures, RBAC, audit trails built in
- **Built for precision shops:** Aerospace, precision machining, sheet metal, medical device manufacturing
- **Strategy + Implementation:** Architect AND deploy through to shop-floor production, not decks
- **ERP/MES integration:** Works with Epicor, Global Shop, E2, ProShop — enhances, doesn't replace
- **Knowledge preservation:** Capture senior estimator/engineer expertise before retirement
- **25+ years operating:** Real business operator mindset for a documentation-heavy industry

ENGAGEMENT MODELS

Fixed-Scope Projects — Defined deliverables, timeline & outcomes

Flexible Consulting — On-demand strategic & implementation support

Fractional AI Advisor — Retained monthly AI leadership & oversight

AI Strategy Workshops — Half-day or full-day executive AI briefings

MANUFACTURING SEGMENTS SERVED

✓ Aerospace & Defense ✓ Precision Machining ✓ Sheet Metal Fabrication
✓ ITAR / CUI Contractors ✓ Industrial Equipment ✓ Electronics & PCB
✓ Automotive Tier 1/2 ✓ Medical Device Mfg

AI TECHNOLOGY STACK

- **LLMs:** Claude (Sonnet/Opus), GPT-4o, Gemini, Whisper
- **Orchestration:** MCP Protocol, Claude Code, Multi-agent systems
- **RAG/Vector:** Weaviate, Chroma, Pinecone, Obsidian.md
- **Platforms:** .NET 8, React, API-first, Multi-tenant SaaS
- **Cloud:** AWS, Azure, Microsoft 365, Private DC

CLIENT IMPACT

Quote Turnaround	Compliance Docs	ITAR Architecture
Days→Hrs	**50%+**	**Air-Gap**
Per job	Less overhead	US-only ready

Ready to put AI to work? — Schedule a consultation: **rjain@technijian.com** | **949.379.8500** | **technijian.com**

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